A FINITE LOCALLY FREE MODULE ON AN INTEGRAL SCHEME CAN BE FILTERED AFTER A MODIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The prefilter flagged this candidate for the phrase 'the lemma is obvious' at proof line 9422; the read shows it dismisses only the rank 0 and rank 1 cases of an induction, so the card is taken under the 2026-10-14 judge-rule (a flagged hit in a long proof is read, not dropped).

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The projective bundle and its structure morphism. Constructions, Section constructions-section-projective-bundle, used at proof lines 9425-9426. Supplies P = P(E) and the canonical surjection.
- The relative Proj is proper. Lemma lemma-relative-proj-proper, used at proof lines 9426-9427. Makes pi proper, hence so is the restriction to X'.
- The scheme theoretic image of a reduced scheme is integral. Morphisms, Lemma morphisms-lemma-scheme-theoretic-image-reduced, used at proof lines 9440-9442. Shows X' is integral.
- Closed immersions and compositions of proper morphisms are proper. Morphisms, Lemmas morphisms-lemma-closed-immersion-proper and morphisms-lemma-composition-proper, used at proof lines 9442-9444. Makes f : X' -> X proper.
- Definition of a modification. Morphisms, Definition morphisms-definition-modification, used at proof lines 9445-9446. Is the notion the constructed f is shown to satisfy.

Source reference. Original section 01OA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2187–2187. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0800: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10165–10165. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 0AAZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13002–13002. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01W5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2470–2470. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01W3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2468–2468. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 056B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6704–6704. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be an integral scheme. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_X$-module. There exists a modification $f : X' \to X$ such that $f^*\mathcal{E}$ has a filtration whose successive quotients are invertible $\mathcal{O}_{X'}$-modules.*

Proof. We prove this by induction on the rank r of $\mathcal{E}$. If $r = 1$ or $r = 0$ the lemma is obvious. Assume $r > 1$. Let $P = \mathbf{P}(\mathcal{E})$ with structure morphism $\pi : P \to X$, see Constructions, Section 01OA. Then π is proper (Lemma 0800). There is a canonical surjection

$$\pi^*\mathcal{E} \to \mathcal{O}_P(1)$$

whose kernel is finite locally free of rank $r-1$. Choose a nonempty open subscheme $U \subset X$ such that $\mathcal{E}|_U \cong \mathcal{O}_U^{\oplus r}$. Then $P_U = \pi^{-1}(U)$ is isomorphic to $\mathbf{P}_U^{r-1}$. In particular, there exists a section $s : U \to P_U$ of π. Let $X' \subset P$ be the scheme theoretic image of the morphism $U \to P_U \to P$. Then X' is integral (Morphisms, Lemma 056B), the morphism $f = \pi|_{X'} : X' \to X$ is proper (Morphisms, Lemmas 01W5 and 01W3), and $f^{-1}(U) \to U$ is an isomorphism. Hence f is a modification (Morphisms, Definition 0AAZ). By construction the pullback $f^*\mathcal{E}$ has a two step filtration whose quotient is invertible because it is equal to $\mathcal{O}_P(1)|_{X'}$ and whose sub $\mathcal{E}'$ is locally free of rank $r - 1$. By induction we can find a modification $g : X'' \to X'$ such that $g^*\mathcal{E}'$ has a filtration as in the statement of the lemma. Thus $f \circ g : X'' \to X$ is the required modification. □